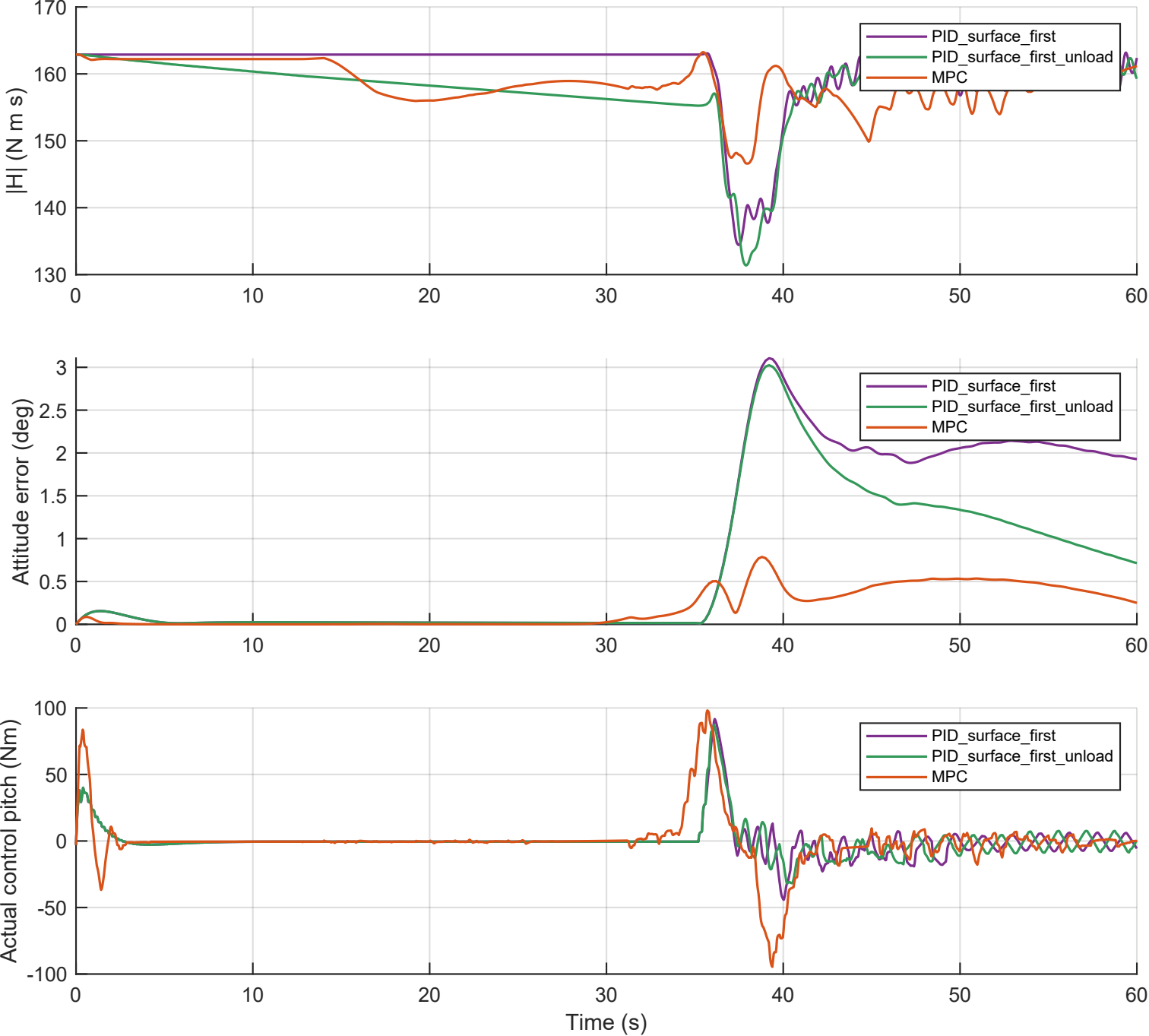

|H| (N m s)
170
160
150
140
130
0
10
20
30
40
50
60
PID_surface_first
PID_surface_first_unload
MPC
Attitude error (deg)
3
2.5
2
1.5
1
0.5
0
0
10
20
30
40
50
60
PID_surface_first
PID_surface_first_unload
MPC
Actual control pitch (Nm)
100
50
0
-50
-100
0
10
20
30
40
50
60
Time (s)
PID_surface_first
PID_surface_first_unload
MPC